

Pharmacies

The science of drugs is known as pharmacology. Pharmacy refers to both the practice of preparing and dispensing drugs, and the place where this happens. A person qualified in pharmacology is called a pharmacist (or chemist). The dispensing chemist can advise on which drugs to use for minor ailments.



Pharmacist at work

Prescription

Some drugs, known as controlled substances, are only available with a doctor's permission. A prescription is a written and signed instruction from a doctor that authorizes a pharmacist to dispense a controlled substance. Prescriptions include the name and dosage of the drug, how often the patient must take it, and any other relevant instructions.



Hospital pharmacy

Over-the-counter drugs

Over-the-counter drugs are available without a prescription. They can be bought at supermarkets and pharmacies, and are usually less powerful than prescription drugs. They have fewer side-effects or contraindications (health problems that warn against their use), but they are still open to misuse. Pharmacists are qualified to recommend certain drug preparations, although they cannot diagnose or prescribe treatment.

Non-medical drugs

Some drugs can be taken for their non-medicinal effects on the mind and body. These effects may include the stimulation or sedation of the mind, a temporary boost to physical performance in sport, or a feeling of emotional well-being.



Wine and brandy

Sedatives

These drugs sedate (slow down) bodily functions, including physical activity and mental agility. Sedatives can make the user feel relaxed and peaceful for a short time. They include sleeping pills, antihistamines (which suppress allergic reactions), antidepressants, and alcohol, which is probably the most widely used non-medical drug in the world.



Stimulants

These drugs temporarily stimulate (speed up) bodily functions and mental processes. However, they can cause after-effects, such as depression. Stimulants include caffeine, nicotine (in tobacco), and cocaine.

Coffee

Drug abuse

This is the improper non-medical use of legal or illegal drugs for physical or psychological reasons. The feelings and mental state experienced by the taker are often very different to that person's actual behaviour, seen by onlookers. After too much alcohol, a drinker may feel bright and witty, while onlookers see a slurring bore.

Customs official arresting a drug trafficker



Illegal drugs

Some drugs are so powerful and dangerous that they are illegal almost everywhere in the world. These include LSD and mescaline (known in some countries as Schedule I drugs), amphetamines, cocaine, and narcotics (Schedule II drugs). Supplying these illegal drugs to users has become a vast international business.



Tobacco shop

Legal drugs

The legality of drugs varies greatly all over the world. As well as the drug's strength and effects, legality often depends on tradition, religion, and availability. One of the most powerful and addictive drugs is alcohol. Alcohol is fully legalized in some countries, partly legalized (for people over 18 or 21) in others, and completely banned in others. Nicotine in the form of cigars, cigarettes, chewing tobacco, and snuff is also legal in most countries.

Group therapy session



Dependence and addiction

A person may come to depend on addictive drugs in order to function. Addiction – intense craving – is hard to control. If the user stops taking the drug, his or her body undergoes "withdrawal", which includes symptoms, such as headaches, sweating, hallucinations, and mood swings. People trying to stop using addictive drugs often find support groups are helpful.

Jonas Salk

Vaccines are substances that give the body resistance or immunity to certain infecting germs. In the 1950s, American microbiologist Jonas Salk (1914–95) developed the first effective vaccine against the crippling disease of polio (poliomyelitis). It spread into worldwide use from 1955 on. From 1960, an oral form of the vaccine, Sabin, gradually replaced the Salk injection.



Timeline

1840s Anaesthetics begin to be used during surgery.

1881 Artificial vaccine used against anthrax.

1910 Paul Ehrlich introduces chemotherapeutic drugs.



1936 Treatment of infections improves with the advent of Protosil, the first sulpha drug.

Fresh witch hazel

1922 Frederick Banting and others treat diabetes using insulin, a natural body hormone.

1940s Howard Florey and Ernst Chain make penicillin available as an antibiotic. It is used widely in World War II.

1956 Oral contraceptives (birth control pills) are introduced, using the natural female hormones, oestrogen and progestogen.



1967 Fertility drugs help couples conceive.

1983 Cyclosporin, an immuno-suppressant, helps prevent rejection of transplanted organs.

1990s AIDS drugs tested.

Tablets and capsules

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